

# Demand: Hicks, Slutsky, and Elasticity

EC306: Intermediate Microeconomics — Day 3

Core reading: Perloff (2020), Ch. 5 and §2.3

# Recap and today's roadmap

**Days 1–2 gave us** the chosen bundle as a function of prices and income: the Marshallian (ordinary) demand  $x^*(p_x, p_y, m)$ , obtained by *maximising utility* subject to a budget.

## Today, in three moves

- ❶ **The dual problem.** Introduce *Hicksian* (compensated) demand and the *expenditure function* — the mirror image of utility maximisation.
- ❷ **Decompose a price change** into a *substitution* and an *income* effect, worked fully with a Cobb–Douglas example (price up *and* down).
- ❸ **Measure responsiveness** precisely with *elasticities*, and link them to revenue.

Hicksian demand is the tool that makes the substitution/income split *exact* rather than hand-wavy.

# Two ways to describe the same consumer

The consumer's problem can be posed from either end.

## Primal — Utility Maximisation (UMP)

“Given prices  $p$  and a *fixed budget*  $m$ , buy the bundle that reaches the *highest utility*.”

$$\max_{x,y} u(x,y) \quad \text{s.t.} \quad p_x x + p_y y = m \quad \implies \quad \underbrace{x^*(p, m)}_{\text{Marshallian demand}}, \quad \underbrace{v(p, m)}_{\text{indirect utility}}.$$

## Dual — Expenditure Minimisation (EMP)

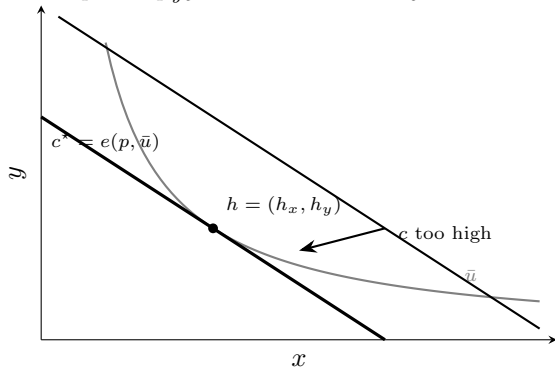
“Given prices  $p$  and a *fixed utility target*  $\bar{u}$ , buy the bundle that reaches it at *least cost*.”

$$\min_{x,y} p_x x + p_y y \quad \text{s.t.} \quad u(x,y) = \bar{u} \quad \implies \quad \underbrace{h(p, \bar{u})}_{\text{Hicksian demand}}, \quad \underbrace{e(p, \bar{u})}_{\text{expenditure function}}.$$

Same tangency condition ( $\text{MRS} = p_x/p_y$ ); different thing held fixed — *income* vs *utility*.

# Expenditure minimisation — the picture

Fix a target indifference curve  $\bar{u}$ . Slide the iso-cost line  $p_x x + p_y y = c$  down until it just touches  $\bar{u}$ :



that tangency is the cheapest way to reach  $\bar{u}$ .

The tangent bundle is the **Hicksian demand**  $h$ ; the height of that lowest iso-cost line is the **expenditure function**  $e(p, \bar{u})$ .

# Hicksian (compensated) demand — definition

## Definition

$h_x(p_x, p_y, \bar{u})$  is the quantity of  $x$  bought when the consumer is *always compensated* to stay on the *same* indifference curve  $\bar{u}$  as prices change.

- It answers: “if a price changes but I keep your *real welfare* (utility) constant, how does your purchase of  $x$  move?”
- Because utility is held fixed, Hicksian demand contains **only the substitution effect** — the income effect has been engineered away by the compensation.
- This is exactly why it is the right tool for the decomposition in Part 2.

## Contrast

Marshallian  $x^*(p, m)$  holds *money income*  $m$  fixed  $\Rightarrow$  contains both effects. Hicksian  $h_x(p, \bar{u})$  holds *utility*  $\bar{u}$  fixed  $\Rightarrow$  substitution only.

# Deriving Hicksian demand — Cobb–Douglas, step by step

Let  $u(x, y) = x^{0.5}y^{0.5}$  and target utility  $\bar{u}$ . Solve the EMP.

## Step 1 — tangency condition

$$\text{MRS} = \frac{MU_x}{MU_y} = \frac{0.5 x^{-0.5} y^{0.5}}{0.5 x^{0.5} y^{-0.5}} = \frac{y}{x} \stackrel{!}{=} \frac{p_x}{p_y} \Rightarrow y = \frac{p_x}{p_y} x.$$

## Step 2 — impose the utility target

$$\text{Substitute into } x^{0.5}y^{0.5} = \bar{u}: \quad x^{0.5} \left( \frac{p_x}{p_y} x \right)^{0.5} = \bar{u} \Rightarrow x \left( \frac{p_x}{p_y} \right)^{0.5} = \bar{u}.$$

## Step 3 — solve for the Hicksian demands

$$h_x(p, \bar{u}) = \bar{u} \left( \frac{p_y}{p_x} \right)^{1/2}, \quad h_y(p, \bar{u}) = \bar{u} \left( \frac{p_x}{p_y} \right)^{1/2}.$$

Note  $h_x$  falls as  $p_x$  rises: the compensated demand curve slopes *down* — always, for every good.

# The expenditure function

## Definition

$$e(p_x, p_y, \bar{u}) = \min_{x, y} \{p_x x + p_y y : u(x, y) = \bar{u}\} = p_x h_x + p_y h_y.$$

For our Cobb–Douglas, plug the Hicksian demands back in:

$$e = p_x \bar{u} \left( \frac{p_y}{p_x} \right)^{1/2} + p_y \bar{u} \left( \frac{p_x}{p_y} \right)^{1/2} = \bar{u} \sqrt{p_x p_y} + \bar{u} \sqrt{p_x p_y} = \boxed{2\bar{u} \sqrt{p_x p_y}}.$$

## Properties every expenditure function has

- **Increasing** in each price and in  $\bar{u}$  (a higher target costs more).
- **Homogeneous of degree 1** in prices: double all prices  $\Rightarrow$  double the cost.
- **Concave** in prices (substitution softens the cost of any single price rise).

# Shephard's lemma — recovering Hicksian demand

## The result

The price-derivative of the expenditure function *is* the Hicksian demand:

$$\frac{\partial e(p, \bar{u})}{\partial p_x} = h_x(p, \bar{u}).$$

**Check with our formulas.** With  $e = 2\bar{u}\sqrt{p_x p_y} = 2\bar{u} p_x^{1/2} p_y^{1/2}$ ,

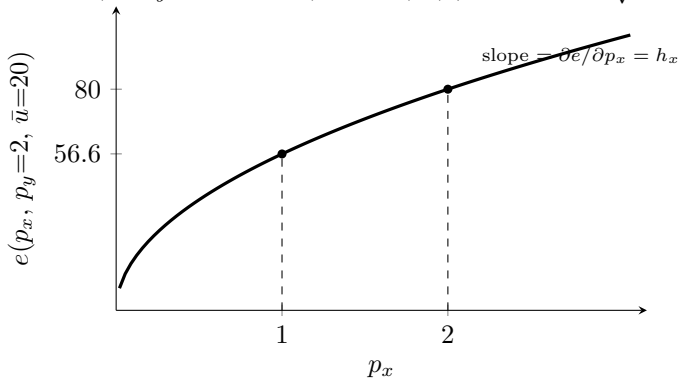
$$\frac{\partial e}{\partial p_x} = 2\bar{u} \cdot \frac{1}{2} p_x^{-1/2} p_y^{1/2} = \bar{u} \left( \frac{p_y}{p_x} \right)^{1/2} = h_x. \quad \checkmark$$

## Why you should care

It means the whole demand system can be read off one scalar function  $e$ . The slope of the expenditure function *in a price direction* tells you exactly how much of that good the compensated consumer buys.

# The expenditure function is concave in price

Plot  $e$  against  $p_x$  (fix  $p_y = 2$ ,  $\bar{u} = 20$ ):  $e = 2(20)\sqrt{2p_x} = 40\sqrt{2}\sqrt{p_x} \approx 56.6\sqrt{p_x}$ .



Concavity  $\Leftrightarrow$  the compensated consumer substitutes away from whatever gets dearer, so cost rises at a *decreasing* rate.

# Marshallian vs Hicksian demand — how they relate

At the *initial* optimum they agree; away from it, Marshallian carries an extra income effect.

The duality identities (evaluate at  $\bar{u} = v(p, m)$ , equivalently  $m = e(p, \bar{u})$ )

$$x^*(p, e(p, \bar{u})) = h_x(p, \bar{u}), \quad h_x(p, v(p, m)) = x^*(p, m).$$

- For a **normal** good the Hicksian curve is *steeper* (less responsive) than the Marshallian: it lacks the reinforcing income effect.
- The horizontal gap between the two curves, at any price away from the start, *is* the income effect.

We draw this gap explicitly in Part 2 — it is the picture of the decomposition.

# Decomposing a price change — the two forces

A fall in  $p_x$  does two things at once.

## Substitution effect (SE)

$x$  is now relatively cheaper  $\Rightarrow$  swap toward  $x$ , *holding utility (real income) fixed*. Always moves *opposite* to the price:  $p_x \downarrow \Rightarrow x \uparrow$ . Unambiguous, for every good. (This is the move *along one indifference curve* — the Hicksian response.)

## Income effect (IE)

A lower  $p_x$  raises *purchasing power*  $\Rightarrow$  the consumer can reach a higher indifference curve. Direction depends on the good:  $\uparrow$  if *normal*,  $\downarrow$  if *inferior*.

Total effect = SE + IE. Hicks holds *utility* fixed; Slutsky holds *the original bundle's affordability* fixed. We use the Hicks version.

# Our worked problem — setup

## Data

$u(x, y) = x^{0.5}y^{0.5}$ , income  $m = 80$ , prices  $p_y = 2$  throughout, and  $p_x$  **falls from 2 to 1**.

Recall the Marshallian demands for this utility (equal exponents  $\Rightarrow$  half of income on each good):

$$x^* = \frac{0.5 m}{p_x} = \frac{40}{p_x}, \quad y^* = \frac{0.5 m}{p_y} = \frac{40}{p_y}.$$

We will find, in four steps:

- 1 the **initial** bundle  $A$ ;
- 2 the **new** bundle  $C$  and the *total* effect;
- 3 the **compensated** bundle  $B$  (Hicksian, holding original utility);
- 4 the **substitution** ( $A \rightarrow B$ ) and **income** ( $B \rightarrow C$ ) effects as numbers.

## Step 1 — the initial bundle $A$

At  $p_x = 2$ ,  $p_y = 2$ ,  $m = 80$ :

$$x_0 = \frac{40}{p_x} = \frac{40}{2} = 20, \quad y_0 = \frac{40}{p_y} = \frac{40}{2} = 20.$$

Initial utility (this is the  $\bar{u}$  we will hold fixed for the substitution step):

$$\bar{u} = u(20, 20) = \sqrt{20 \cdot 20} = 20.$$

### Bundle $A$

$$A = (x_0, y_0) = (20, 20), \quad \bar{u} = 20.$$

Spending check:  $2(20) + 2(20) = 80 = m$ . ✓

## Step 2 — the new bundle $C$ and the total effect

Now  $p_x = 1$  (with  $p_y = 2$ ,  $m = 80$  unchanged):

$$x_1 = \frac{40}{p_x} = \frac{40}{1} = 40, \quad y_1 = \frac{40}{p_y} = \frac{40}{2} = 20.$$

### Bundle $C$ and total effect

$$C = (x_1, y_1) = (40, 20), \quad \Delta x_{\text{total}} = x_1 - x_0 = 40 - 20 = +\mathbf{20}.$$

New utility has risen:  $u(40, 20) = \sqrt{40 \cdot 20} = \sqrt{800} \approx 28.3 > 20$ . The consumer is better off — that welfare gain is what the income effect will capture.

## Step 3 — the compensated bundle $B$ (Hicksian)

Hold utility at  $\bar{u} = 20$ , but face the *new* price  $p_x = 1$ . Use the Hicksian demands derived earlier:

$$h_x = \bar{u} \left( \frac{p_y}{p_x} \right)^{1/2} = 20 \left( \frac{2}{1} \right)^{1/2} = 20\sqrt{2} \approx \mathbf{28.28}, \quad h_y = \bar{u} \left( \frac{p_x}{p_y} \right)^{1/2} = 20 \left( \frac{1}{2} \right)^{1/2} = 10\sqrt{2} \approx 14.14.$$

### Bundle $B$ and the compensating expenditure

$$B = (x_c, y_c) = (28.28, 14.14), \quad e(1, 2, 20) = 2\bar{u}\sqrt{p_x p_y} = 2(20)\sqrt{2} \approx \mathbf{56.57}.$$

$B$  is the cheapest way to keep utility = 20 at the *new* prices; it needs only \$56.57, not \$80. The “extra” \$23.43 of real purchasing power is the income effect.

## Step 4 — read off the two effects

$$\underbrace{x_0 = 20}_A \xrightarrow{\text{SE}} \underbrace{x_c = 28.28}_B \xrightarrow{\text{IE}} \underbrace{x_1 = 40}_C$$

Substitution effect (move along the original indifference curve)

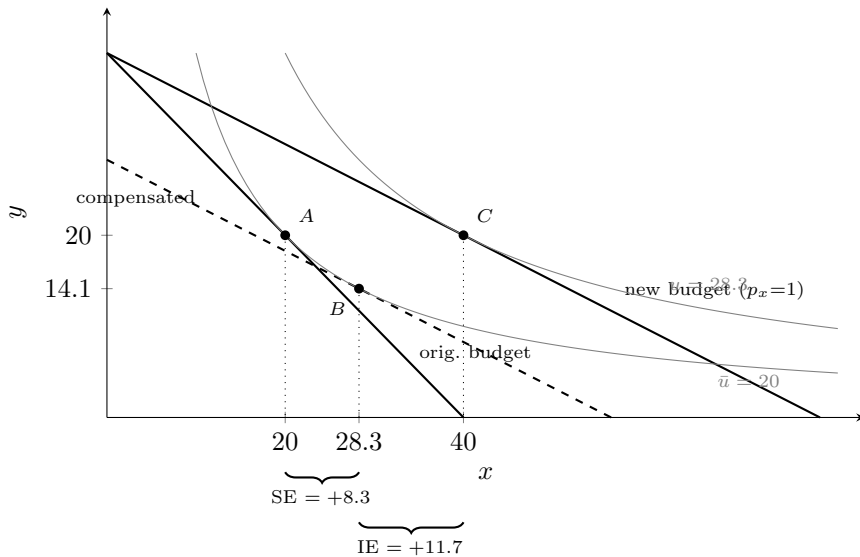
$$\text{SE} = x_c - x_0 = 28.28 - 20 = +\mathbf{8.28}.$$

Income effect (jump to the higher indifference curve)

$$\text{IE} = x_1 - x_c = 40 - 28.28 = +\mathbf{11.72}.$$

Check:  $\text{SE} + \text{IE} = 8.28 + 11.72 = 20 = \Delta x_{\text{total}}$ . ✓ Both positive  $\Rightarrow x$  is a **normal** good.

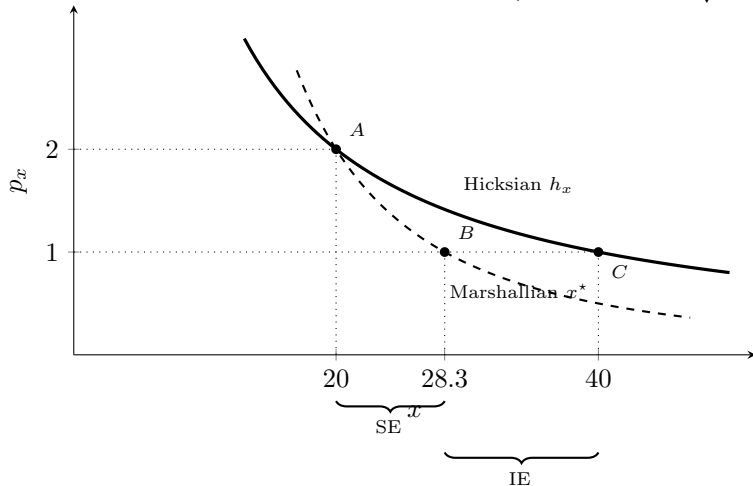
# The Slutsky–Hicks picture (in goods space)



$A \rightarrow B$ : slide along the *old* curve to the new slope (substitution).  $B \rightarrow C$ : jump out to the higher curve (income). 17/31

# The same decomposition on the demand curve

Plot  $x$  against  $p_x$ . Marshallian  $x^* = 40/p_x$ ; Hicksian  $h_x = 20\sqrt{2/p_x} = \sqrt{800}/\sqrt{p_x}$ . They cross at  $A$ .



At  $p_x = 1$ : the Hicksian gives  $x = 28.3$  (substitution only); the Marshallian gives  $x = 40$ . The horizontal gap  $28.3 \rightarrow 40$  is the income effect.

## The mirror case — a price *increase*

Reverse the experiment:  $p_x$  rises from 1 to 2 (now  $A = (40, 20)$ ,  $C = (20, 20)$ , and we compensate to  $\bar{u} = u(40, 20) = 28.3$ ).

Hicksian at the new higher  $p_x = 2$ , holding  $\bar{u} = 28.3$ :

$$h_x = 28.3\sqrt{\frac{2}{2}} = 28.3.$$

So  $x : 40 \xrightarrow{\text{SE}} 28.3 \xrightarrow{\text{IE}} 20$ .

- $\text{SE} = 28.3 - 40 = -11.7$  (dearer  $\Rightarrow$  substitute away: negative, always).
- $\text{IE} = 20 - 28.3 = -8.3$  (poorer in real terms;  $x$  normal  $\Rightarrow$  also negative).

### Lesson

For a **normal** good, SE and IE point the *same* way, so demand slopes down unambiguously. Sign of SE never changes; only the IE's sign (and size) can, if the good is inferior.

# The Slutsky equation — the algebra behind the picture

The decomposition is an exact identity:

$$\underbrace{\frac{\partial x^*}{\partial p_x}}_{\text{total (Marshallian)}} = \underbrace{\frac{\partial h_x}{\partial p_x}}_{\text{substitution } (\leq 0)} - \underbrace{x^* \frac{\partial x^*}{\partial m}}_{\text{income}}$$

- The substitution term is *always*  $\leq 0$  (compensated demand slopes down).
- The income term carries the sign of  $\partial x^* / \partial m$ : subtract a *positive* for a normal good (reinforces the fall), a *negative* for an inferior good (offsets it).

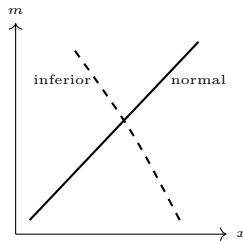
## Giffen, precisely

Demand slopes *up* ( $\partial x^* / \partial p_x > 0$ ) only if the good is inferior *and* the income term outweighs the substitution term. That is the entire content of “Giffen”.

# Engel curves: how demand responds to income

Holding prices fixed, raise  $m$  and plot  $x^*$  against  $m$ : the **Engel curve**. This is the “income” margin of the last section, isolated.

- **Normal good:**  $\partial x^* / \partial m > 0$ . (Cobb–Douglas:  $x^* = 0.5m/p_x$  — a straight line through the origin.)
- **Inferior good:**  $\partial x^* / \partial m < 0$  over some range — demand *falls* as you get richer.



## Real inferior goods

Bus travel, instant noodles, supermarket own-brand staples: as income rises, people substitute toward cars, restaurants, and premium brands. Inferiority is about *position in the budget*, not quality.

# Giffen goods: when the income effect wins

For a Giffen good the negative income effect (it is strongly inferior) *outweighs* the substitution effect, so demand **rises** when price rises. Note our Cobb–Douglas can *never* do this:  $x^* = 0.5m/p_x$  is strictly increasing in  $m$ , so it is always normal.

## The one credible empirical case

Jensen & Miller (2008) found suggestive Giffen behaviour for rice/wheat among very poor households in China: when the staple got dearer, families dropped pricier protein and ate *more* of the staple to hit calorie needs. Requires a good that is (i) strongly inferior, (ii) a large budget share, (iii) few substitutes. Vanishingly rare — but theoretically real, not a paradox.

**Exam trap:** “downward-sloping demand” is a *result*, not an axiom. It can fail. Know exactly when.

# Own-price elasticity of demand

$$\varepsilon = \frac{\% \Delta Q}{\% \Delta p} = \frac{\partial Q}{\partial p} \cdot \frac{p}{Q}.$$

A *unit-free* measure — the reason we don't just use the slope  $\partial Q / \partial p$  (which depends on units: litres vs gallons would change it, elasticity is immune).

- $|\varepsilon| > 1$ : **elastic** (quantity reacts strongly — luxuries, many substitutes).
- $|\varepsilon| < 1$ : **inelastic** (quantity barely moves — necessities, addictive goods).
- $|\varepsilon| = 1$ : unit elastic.

## Numbers from the real world

Short-run petrol demand  $\approx -0.25$  (inelastic: you still commute). Cigarettes  $\approx -0.4$ . Restaurant meals or a specific airline route  $\approx -2$  to  $-4$  (elastic: easy substitutes). These signs and magnitudes drive tax and pricing policy.

# Point vs arc elasticity — and a common error

**Point** (calculus, one price):  $\varepsilon = \frac{dQ}{dp} \frac{p}{Q}$ .

**Arc** (between two observed points, using midpoints):

$$\varepsilon_{\text{arc}} = \frac{(Q_2 - Q_1)/[(Q_1 + Q_2)/2]}{(p_2 - p_1)/[(p_1 + p_2)/2]}.$$

## Why the midpoint formula exists

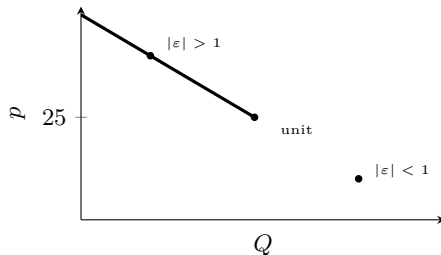
Naive  $\% \Delta$  gives different answers depending on which point is the “base”. The midpoint method is symmetric, so a rise from  $10 \rightarrow 12$  and a fall from  $12 \rightarrow 10$  give the same elasticity. Use arc for discrete data, point for a known demand function.

## Worked example — linear demand has varying elasticity

$$Q = 100 - 2p.$$

$$\frac{dQ}{dp} = -2, \quad \varepsilon = -2 \cdot \frac{p}{100 - 2p}.$$

| $p$ | $Q$ | $\varepsilon$ | region       |
|-----|-----|---------------|--------------|
| 10  | 80  | -0.25         | inelastic    |
| 25  | 50  | -1.00         | unit elastic |
| 40  | 20  | -4.00         | elastic      |



### Key lesson

A *straight-line* demand curve does **not** have constant elasticity. Slope is constant ( $-2$ ); elasticity runs from  $0$  at the bottom to  $-\infty$  at the top, passing through  $-1$  at the midpoint.

# Constant-elasticity demand

Contrast:  $Q = Ap^{-\eta}$  has the *same* elasticity everywhere.

$$\varepsilon = \frac{dQ}{dp} \frac{p}{Q} = A(-\eta)p^{-\eta-1} \cdot \frac{p}{Ap^{-\eta}} = -\eta \quad (\text{for all } p).$$

## Cobb–Douglas is the unit-elastic special case

Our Marshallian  $x^* = 40/p_x = 40p_x^{-1}$  has  $\eta = 1$ : own-price elasticity  $\equiv -1$  at every price. That is a *hallmark* of Cobb–Douglas — constant budget shares mean spending on  $x$  never changes, so  $p_x \uparrow$  by 1% cuts  $x$  by exactly 1%.

A log-log plot of constant-elasticity demand is a straight line with slope  $-\eta$ .

# Elasticity and total revenue

$TR = p \cdot Q$ . Differentiating,  $\frac{dTR}{dp} = Q(1 + \varepsilon)$ . So:

- Inelastic ( $|\varepsilon| < 1$ ): raise price  $\Rightarrow$  revenue **rises**.
- Elastic ( $|\varepsilon| > 1$ ): raise price  $\Rightarrow$  revenue **falls**.
- Revenue is maximised at  $|\varepsilon| = 1$  (and for Cobb–Douglas, revenue on  $x$  is *constant* —  $\varepsilon \equiv -1$ ).

## Why governments tax cigarettes and petrol

Inelastic demand means a tax raises lots of revenue with little quantity drop — and the consumer bears most of the burden. The same logic tells a streaming firm whether a price rise grows or shrinks revenue: it hinges entirely on where  $|\varepsilon|$  sits relative to 1.

# Cross-price and income elasticities

**Cross-price:**  $\varepsilon_{xy} = \frac{\partial Q_x}{\partial p_y} \frac{p_y}{Q_x}$ .

- $> 0$ : **substitutes** (Pepsi vs Coke).
- $< 0$ : **complements** (printers vs ink).
- Cobb–Douglas:  $\varepsilon_{xy} = 0$  ( $x^*$  has no  $p_y$ ).

**Income:**  $\varepsilon_m = \frac{\partial Q}{\partial m} \frac{m}{Q}$ .

- $> 1$ : luxury.  $0 < \varepsilon_m < 1$ : necessity.
- $< 0$ : inferior good.
- Cobb–Douglas:  $\varepsilon_m = +1$ .

## Competition policy uses cross-price elasticity directly

Whether two products are “in the same market” for a merger case is, operationally, a cross-price-elasticity question: if a price rise in A sharply lifts demand for B, they constrain each other.

# Your turn (1) — in-lecture

## Your turn

Demand is  $Q = 240 - 4p$ .

- 1 Find the price at which demand is unit elastic.
- 2 A firm currently charges  $p = 20$ . Should it raise or cut price to increase revenue? Justify with the elasticity.

## Your turn (2) — in-lecture

### Your turn

A consumer has  $u = x^{0.5}y^{0.5}$ ,  $m = 100$ ,  $p_y = 1$ , and  $p_x$  **ris**es from 1 to 2.

- ① Find the initial and final bundles and the total change in  $x$ .
- ② Using the Hicksian demands  $h_x = \bar{u}\sqrt{p_y/p_x}$ , find the compensated bundle and split  $\Delta x$  into substitution and income effects.

# What you can now do

- ➊ State the dual problem: derive **Hicksian demand** and the **expenditure function**, and use Shephard's lemma.
- ➋ Decompose a price change into **substitution** and **income** effects — numerically (Cobb–Douglas) and in both the goods-space and demand-curve diagrams.
- ➌ Read the **Slutsky equation** and say exactly when demand could slope up (Giffen).
- ➍ Compute and interpret own-price, cross-price, and income **elasticities**, and link them to total revenue and to real tax/pricing/competition decisions.

## Tomorrow

We switch sides of the market. Same optimisation logic — but now the **firm** choosing inputs to make output. Production functions, marginal product, and returns to scale.